

Supplementary material: HCA-DSE

Full budget sweeps and validation subsets accompanying the main manuscript. All values are generated from the same result CSVs as the main-text summaries.

Table S1: Full budget-matched comparison on S2. Medians over 20 seeds.							
scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S2	telemetry	exhaustive	3 888	1.000	1.000	0.0000	0.060
		random ($B=100$)	100	0.037	0.692	0.1496	0.001
		random ($B=250$)	250	0.037	0.794	0.1018	0.004
		random ($B=500$)	500	0.074	0.880	0.0713	0.008
		random ($B=1000$)	1000	0.185	0.923	0.0451	0.017
		NSGA-II ($B=100$)	100	0.037	0.703	0.1537	0.014
		NSGA-II ($B=250$)	250	0.259	0.886	0.0447	0.037
		NSGA-II ($B=500$)	500	0.759	0.995	0.0063	0.084
		NSGA-II ($B=1000$)	1000	0.944	0.999	0.0003	0.184
		HCA-DSE	50	1.000	1.000	0.0000	0.017
S2	control	exhaustive	3 888	1.000	1.000	0.0000	0.050
		random ($B=100$)	100	0.040	0.691	0.1460	0.001
		random ($B=250$)	250	0.040	0.792	0.0940	0.004
		random ($B=500$)	500	0.080	0.879	0.0566	0.007
		random ($B=1000$)	1000	0.160	0.923	0.0379	0.017
		NSGA-II ($B=100$)	100	0.040	0.627	0.1592	0.014
		NSGA-II ($B=250$)	250	0.300	0.910	0.0348	0.039
		NSGA-II ($B=500$)	500	0.800	0.991	0.0041	0.091
		NSGA-II ($B=1000$)	1000	0.900	0.999	0.0007	0.197
		HCA-DSE	39	1.000	1.000	0.0000	0.015
S2	sensing	exhaustive	3 888	1.000	1.000	0.0000	0.035
		random ($B=100$)	100	0.000	0.469	0.2629	0.001
		random ($B=250$)	250	0.062	0.686	0.1521	0.003
		random ($B=500$)	500	0.062	0.804	0.0808	0.006
		random ($B=1000$)	1000	0.188	0.866	0.0622	0.012
		NSGA-II ($B=100$)	100	0.000	0.404	0.3482	0.014
		NSGA-II ($B=250$)	250	0.312	0.793	0.0782	0.041
		NSGA-II ($B=500$)	500	0.719	0.965	0.0224	0.102
		NSGA-II ($B=1000$)	1000	0.938	1.000	0.0009	0.235
		HCA-DSE	26	1.000	1.000	0.0000	0.012

Table S2: Full budget-matched comparison on S3. Medians over 20 seeds.

scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S3	telemetry	exhaustive	34 560	1.000	1.000	0.0000	0.354
		random ($B=100$)	100	0.000	0.259	0.3903	0.001
		random ($B=250$)	250	0.000	0.487	0.2574	0.003
		random ($B=500$)	500	0.000	0.563	0.1671	0.006
		random ($B=1000$)	1000	0.016	0.699	0.1147	0.013
		NSGA-II ($B=100$)	100	0.000	0.151	0.4919	0.012
		NSGA-II ($B=250$)	250	0.000	0.687	0.1732	0.033
		NSGA-II ($B=500$)	500	0.355	0.915	0.0558	0.078
		NSGA-II ($B=1000$)	1000	0.645	0.987	0.0415	0.199
		HCA-DSE	56	1.000	1.000	0.0000	0.063
S3	control	exhaustive	34 560	1.000	1.000	0.0000	0.323
		random ($B=100$)	100	0.000	0.262	0.4192	0.001
		random ($B=250$)	250	0.000	0.492	0.2537	0.003
		random ($B=500$)	500	0.000	0.564	0.1746	0.006
		random ($B=1000$)	1000	0.018	0.691	0.1248	0.012
		NSGA-II ($B=100$)	100	0.000	0.286	0.4405	0.015
		NSGA-II ($B=250$)	250	0.037	0.748	0.1009	0.038
		NSGA-II ($B=500$)	500	0.278	0.933	0.0387	0.086
		NSGA-II ($B=1000$)	1000	0.648	0.991	0.0244	0.209
		HCA-DSE	43	1.000	1.000	0.0000	0.065
S3	sensing	exhaustive	34 560	1.000	1.000	0.0000	0.356
		random ($B=100$)	100	0.000	0.037	0.6082	0.001
		random ($B=250$)	250	0.000	0.231	0.3843	0.003
		random ($B=500$)	500	0.000	0.556	0.2543	0.006
		random ($B=1000$)	1000	0.000	0.600	0.1747	0.012
		NSGA-II ($B=100$)	100	0.000	0.050	0.5191	0.014
		NSGA-II ($B=250$)	250	0.037	0.279	0.2833	0.035
		NSGA-II ($B=500$)	500	0.370	0.750	0.1996	0.082
		NSGA-II ($B=1000$)	1000	0.741	0.969	0.1670	0.218
		HCA-DSE	43	1.000	1.000	0.0000	0.074

Table S3: Full S3 statistical comparison: one-sided Mann–Whitney test for larger NSGA-II hypervolume and Cliff’s δ . Unadjusted p -values.

workload	budget	NSGA-II HV med.	random HV med.	p	Cliff’s δ	NSGA-II recall	HCA-DSE evals
telemetry	100	0.1509	0.259	0.951	-0.302	0.0	56
telemetry	250	0.6868	0.4871	0.021	0.38	0.0	56
telemetry	500	0.9154	0.5633	<0.001	0.7	0.355	56
telemetry	1000	0.9874	0.6993	<0.001	0.7	0.645	56
control	100	0.2859	0.2616	0.484	0.01	0.0	43
control	250	0.7481	0.4916	<0.001	0.62	0.037	43
control	500	0.9331	0.5644	<0.001	0.8	0.278	43
control	1000	0.9906	0.6913	<0.001	0.8	0.648	43
sensing	100	0.0497	0.0367	0.261	0.12	0.0	43
sensing	250	0.2787	0.2308	0.672	-0.08	0.037	43
sensing	500	0.7504	0.5561	0.168	0.18	0.37	43
sensing	1000	0.9689	0.5998	0.054	0.3	0.741	43

Table S4: Testbed validation against measured p_{50} , including utilisation and reproducibility subsets. Pair agreement is a fraction; undefined values (no eligible pairs) are shown as –.

subset	n	MAPE [%]	med. AE [ms]	Spearman ρ	Kendall τ	pair > 10%	pair > 25%	repeat IQR [ms]
telemetry (all)	12	20.3	7.54	0.839	0.727	0.852	0.895	0.23
telemetry ($\rho \leq 0.75$)	6	16.2	2.91	0.829	0.733	0.818	1.0	0.18
control (all)	12	15.7	2.56	0.872	0.779	0.917	0.909	0.27
control ($\rho \leq 0.75$)	10	9.5	2.4	0.778	0.674	0.852	0.833	0.17
sensing (all)	12	50.0	20.0	0.949	0.87	0.957	0.957	2.17
sensing ($\rho \leq 0.75$)	10	49.7	19.28	0.985	0.944	1.0	1.0	1.86
sensing (reproducible)	8	41.7	18.77	0.994	0.982	1.0	–	1.49
all workloads (all)	36	28.7	7.54	0.97	0.893	0.968	0.982	0.49
all workloads ($\rho \leq 0.75$)	26	26.5	3.91	0.948	0.861	0.962	0.982	0.4
all workloads (reproducible)	32	23.9	4.54	0.96	0.881	0.963	0.979	0.38

Table S5: Full order/queue ablation on S1–S3. All configurations preserve the unrounded exhaustive front.

scale	workload	order	queue	states	checks	evals	active	pruned	ms
S1	telemetry	determinants-early	off	447	746	136	0	0	10.0
S1	telemetry	determinants-early	on	251	516	34	306	46	7.9
S1	telemetry	determinants-late	off	695	1299	136	0	0	15.8
S1	telemetry	determinants-late	on	591	1299	32	0	0	17.3
S1	control	determinants-early	off	391	669	113	0	0	9.1
S1	control	determinants-early	on	242	512	28	312	35	8.0
S1	control	determinants-late	off	679	1317	112	0	0	15.5
S1	control	determinants-late	on	593	1317	26	0	0	17.7
S1	sensing	determinants-early	off	481	810	147	0	0	9.5
S1	sensing	determinants-early	on	226	503	16	309	72	6.1
S1	sensing	determinants-late	off	497	819	148	0	0	10.1
S1	sensing	determinants-late	on	365	819	16	0	0	8.6
S2	telemetry	determinants-early	off	939	1815	208	0	0	25.4
S2	telemetry	determinants-early	on	608	1383	50	839	99	22.0
S2	telemetry	determinants-late	off	1816	3940	208	0	0	50.7
S2	telemetry	determinants-late	on	1656	3940	48	0	0	57.5
S2	control	determinants-early	off	835	1674	165	0	0	22.3
S2	control	determinants-early	on	587	1364	39	842	80	20.7
S2	control	determinants-late	off	1780	3960	164	0	0	49.1
S2	control	determinants-late	on	1653	3960	37	0	0	55.4
S2	sensing	determinants-early	off	1161	2184	289	0	0	26.9
S2	sensing	determinants-early	on	600	1427	26	879	177	17.4
S2	sensing	determinants-late	off	1268	2421	290	0	0	28.8
S2	sensing	determinants-late	on	1004	2421	26	0	0	26.8
S3	telemetry	determinants-early	off	2871	6806	251	0	0	73.3
S3	telemetry	determinants-early	on	2401	6100	56	3777	188	71.8
S3	telemetry	determinants-late	off	6029	14794	251	0	0	169.5
S3	telemetry	determinants-late	on	5832	14794	54	0	0	205.1
S3	control	determinants-early	off	2755	6636	206	0	0	65.9
S3	control	determinants-early	on	2375	6074	43	3777	163	78.6
S3	control	determinants-late	off	5990	14812	205	0	0	155.6
S3	control	determinants-late	on	5826	14812	41	0	0	174.0
S3	sensing	determinants-early	off	4351	9905	537	0	0	107.9
S3	sensing	determinants-early	on	3221	8281	43	5139	419	86.2
S3	sensing	determinants-late	off	4888	11367	538	0	0	103.4
S3	sensing	determinants-late	on	4393	11367	43	0	0	109.4